$$\mathrm{f}(x) = x^3 + (p + 3)\,x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $\mathrm{f}(x)$

(a) show that

$$9p + q = -51$$

(2)

Given also that when $\mathrm{f}(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0$$

(2)

(c) Hence find the value of p and the value of q.

(3)

(d) Hence find a quadratic expression $\mathrm{g}(x)$ such that

$$\mathrm{f}(x) = (x - 3)\,\mathrm{g}(x)$$

(2)